pillar-UDP Redundancy: Header Format and the Loss/Bandwidth Trade

(companion to "Clustered Multipath Congestion Avoidance and Redundant Delivery")

pillar engineering

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Abstract

This is a working draft. The primary pillar-UDP paper proves the *qualitative* delivery guarantees (exactly-once, route-around liveness, bounded fan-out) and defers two quantitative matters to this companion: (i) the on-the-wire encoding of the per-connection *redundancy allowance* and the rest of the pillar-UDP message header, and (ii) the analytic loss/bandwidth trade that determines how large the allowance should be for a target reliability. This draft fixes the analytic framework—which is closed-form and stated here in full—and lays out the header's logical field set; the exact byte encoding and the empirical validation of the independence assumption remain open (§5).

1 Scope

The companion's job is narrow: give operators a defensible rule for setting the redundancy allowance, and give implementers a stable header contract. Nothing here changes the protocol's proven guarantees; it parameterises them. Two regimes share one substrate and differ only in redundancy shape: *small-message* delivery (a message rides in W whole redundant copies) and *bulk* delivery (a payload is split into K data shards plus M erasure shards and reconstructed from any K arrivals). Both are analysed below.

2 The loss/bandwidth trade (small messages)

Let a message be sent as W redundant copies over paths whose losses are independent with per-path probability p (the primary paper argues dispersion across AS-diverse nodes is what makes independence realistic; §5 records this as the assumption to validate empirically). The message is undelivered only if every copy drops:

$$P_{\text{fail}}(W, p) = p^W.$$

The bandwidth cost is a fan-out of W: overhead is $(W-1)$ redundant copies per delivered message. Fixing a target residual ε and solving $p^W \le \varepsilon$ gives the *minimum allowance*

$$W^\star(p, \varepsilon) = \left\lceil \frac{\ln \varepsilon}{\ln p} \right\rceil = \left\lceil \log_p \varepsilon \right\rceil,$$

p	ε=10^{-3}	ε=10^{-6}	ε=10^{-9}
0.01	2	3	5
0.05	3	5	7
0.10	3	6	9
0.20	5	9	13
0.30	6	12	18
0.50	10	20	30

Table 1: Minimum redundancy allowance $W^\star = \lceil \log_p \varepsilon \rceil$ for target residual loss ε. The allowance is cheap on good paths and grows only as the path degrades.

a closed form the sender can evaluate per connection from its measured p and the connection's declared ε. This is the value the redundancy allowance encodes; it scales *down* toward $W = 1$ as $p \to 0$ (a clean path costs nothing) and *up* as $p \to 1$, which is exactly the adaptive posture the primary paper requires. Table 1 tabulates $W^\star$.

3 Erasure regime (bulk)

For bulk transfer the payload is split into K data shards and M parity shards ($N = K + M$ total, systematic code), reconstructable from any K arrivals (the implementation's `encode`/`reconstruct` in `pillar_udp.rs`). With independent per-shard loss p, reconstruction fails iff fewer than K of N shards arrive:

$$P_{\text{fail}}(K, M, p) = \sum_{i=0}^{K-1} \binom{N}{i} (1-p)^i p^{N-i}.$$

The bandwidth overhead is M/K, tunable independently of K, so bulk pays a *fractional* overhead where small messages pay a multiple. The parity rate M/K is tied to the same redundancy-allowance header field, mapped through the ε/p relation above rather than a raw copy count. **[TODO: plot $P_{\text{fail}}(K, M, p)$ surface and derive the M/K that meets a given ε at measured p; cross-check against Table 1.]**

4 Header format (logical fields)

The header binds every datagram—initial spray copy or forward—to its logical message and states the connection's redundancy contract up front. The *logical* field

field	meaning
CID	content address of the message bytes; the dedup and integrity key (identical to the streamdb/blob content address).
redundancy allowance	the connection's $W^\star$ (small) or parity rate M/K (bulk), per §2–3; caps system-wide fan-out.
hop TTL	remaining forward budget; decremented per hop, forwarding stops at 0 (with CID dedup, guarantees loop-free termination).
side-effect class	idempotent/*Convergent* vs *Exclusive*; routes an exclusive message's processing to the coordination-core lease holder.
shard descriptor	for bulk: K, M, and the shard index/offset for reconstruction.
return-routability token	anti-amplification: proof the source address is reachable before the cell commits redundant replies.

Table 2: pillar-UDP header logical fields. Each corresponds to an invariant already modelled in `specs/PillarUDP.tla` / `PillarUdpClient.tla` / `PillarUdpMesh.tla`; this paper's remaining work is their wire encoding.

set below is fixed by the protocol's proven mechanics; the concrete byte layout is deferred (§5).

5 Open items

This draft is complete on the analytic framework and the logical header contract; the following remain before it is publishable:

- **[TODO: Byte-level wire encoding of Table 2: field widths, varint vs fixed, alignment, and the Noise/session-key framing boundary.]**
- **[TODO: Empirical validation of the independence assumption (§2): measure correlated loss across AS-diverse pillar-ipam prefixes and quantify how much it inflates the effective p used in $W^\star$.]**
- **[TODO: Bulk M/K selection curve (§3) and its mapping onto the single redundancy-allowance field shared with the small-message regime.]**
- **[TODO: Congestion-safety envelope: bound the aggregate fan-out a cell emits so the adaptive allowance cannot itself induce congestion collapse under correlated loss.]**

No quantitative claim in this draft rests on unmeasured data: Tables 1 is exact arithmetic from the closed forms, and every empirical question is listed above rather than filled with a placeholder value.